MLFF-DATA8: MACE Artifacts, Replay, and Protocol Identity

mdstats project

2026-07-28

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Status and scope

MLFF-DATA8 converts DATA5-DATA7 evidence into executable, fixed-file MACE job artifacts. It does not run training, choose a checkpoint, activate a locked test, or perform active learning. The first adapter is intentionally narrow and is locked to `mace-torch==0.3.16` with the `NATIVE_MACE_FIXED` exposure backend.

The stage owns five boundaries:

1. MACE-readable target extended XYZ plus compact sidecar provenance;
2. explicit atomic-reference mappings and target training weights;
3. replay train/monitor preparation with disjointness evidence;
4. complete, immutable `TrainingProtocolIdentity` records;
5. independent final-development and cross-validation job directories in which held-out evaluation and locked interpolation test data never appear in a training configuration.

DATA8 is an artifact-preparation gate. DATA9 owns process execution, candidate checkpoint evaluation, replay-retention enforcement, out-of-fold aggregation, final committee construction, and protocol freeze.

Why the adapter is version locked

MACE is an evolving training application rather than a static file format. The adapter therefore verifies the exact source behavior on which its artifacts depend. For MACE v0.3.16, the required behaviors are:

- the replay head `pt_head` is sorted before target heads;
- validation heads are evaluated in loader order;
- native scheduler, patience, and best-checkpoint logic use only the last validation head;
- target data can be duplicated internally when the target/replay ratio falls below `real_pt_data_ratio_threshold`;
- `dry_run` and `save_all_checkpoints` are available;
- external replay supports `pt_train_file` and `pt_valid_file`.

The first adapter uses target-last ordering so native scheduling observes the target monitor, disables implicit duplication by writing `real_pt_data_ratio_threshold: 0.0`, requests all candidate checkpoints, and records a loader dry-run prediction. DATA9 later applies the declared target, cation-resolved, stress, and replay-retention constraints externally.

A source probe is evidence, not a claim that any future MACE version is compatible. Every supported version requires its own tested compatibility record.

Public contracts

`MaceCompatibilityPolicy`

Locks package name, package version, release tag, commit, official source URLs, and policy version. The initial policy accepts only `mace-torch==0.3.16`.

`MaceSourceProbe`

Records content digests and verified source semantics:

- replay-head ordering;
- target-last validation ordering;
- last-head checkpoint behavior;
- implicit target duplication;
- dry-run support;

- save-all-checkpoint support;
- fixed-file adapter acceptance.

The adapter fails closed if any required behavior is absent.

`MaceCheckpointControlPolicy`

The initial mode is `NATIVE_TARGET_LAST_WITH_EXTERNAL_CONSTRAINT_AUDIT`. It requires:

- target validation head last;
- all candidate checkpoints saved;
- native early stopping effectively neutralized by large patience;
- external replay-retention audit when replay is enabled;
- no locked-test evidence in checkpoint selection.

`MaceLoaderDryRun`

Predicts the realized loader contract before training:

- head and validation-head order;
- native checkpoint head;
- requested and effective target/replay counts;
- implicit target duplication factor;
- target/replay ratio;
- fixed-file backend identity.

For target count N_{ft}, replay count N_{pt}, and threshold r, the v0.3.16 compatibility emulation repeats target data until the realized ratio satisfies the source behavior. DATA8 defaults to $r = 0$, so balancing is owned explicitly by mdstats rather than hidden inside MACE.

`MaceExtxyzArtifact`

Stores one verified target data file and its sidecar manifest. Every exported configuration contains the minimum MACE-readable keys:

```
REF_energy
REF_forces
REF_stress
config_weight
config_energy_weight
config_forces_weight
config_stress_weight
frame_uid
```

The exporter performs an ASE write/read round trip and verifies frame order, labels, weights, and stress representation. Per-atom floating columns are written with at least 17 significant decimal digits. The ASE 3.29 default `%16.8f` format is not used because it can round Cartesian positions and force labels by several nanounits and violate the lossless DATA8 contract.

Executable loss family and weighting layers

Every generated MACE configuration on a current path - target-size candidate training, post-selection cross-validation, and fresh final production - SHALL emit the resolved global objective coefficients explicitly:

```
loss = "stress"
energy_weight
forces_weight
stress_weight
```

`loss = "stress"` selects pinned MACE's `WeightedEnergyForcesStressLoss`. Its native reductions consume `config_weight` (`ref.weight`) and the per-frame property weights **linearly**, and apply the global coefficients exactly once, outside those reductions. No current path may rely on MACE's `forces_weight = 100` default.

MACE's `UniversalLoss` SHALL NOT be used on a current path. Its per-config property weights scale residuals *inside* a Huber evaluation, so they are not linearly equivalent to global objective coefficients, and it does not consume `config_weight` at all

- it therefore cannot realize the declared mdstats weighting contract. This SHALL NOT be worked around with a patched loss, a square-root weighting trick, residual pre-scaling, sample duplication, or a second mdstats loss engine.

The three weighting layers are distinct owners:

Layer	Owner	Exported as
global loss coefficients	`[objective]` / `TrainingObjectivePolicy`	`energy_weight`, `forces_weight`, `stress_weight` config keys
per-configuration weight	`[weighting]` / `ConfigurationWeightPolicy`	`config_weight`
local property weights	canonical label presence	`config_energy_weight`, `config_forces_weight`, `config_stress_weight`

Local property weights are availability masks - `1.0` when the canonical label is present, `0.0` when it is absent - and SHALL NOT carry per-frame copies of the global coefficient ratio.

The loss family is part of model/training-method identity. Model reconstruction records it, and pinned MACE derives the same `compute_stress` / `compute_virials` output configuration for it as for the retired family, so reconstruction and EVAL2 semantics are preserved while the identity now names what actually executes. Checkpoints produced under the retired loss semantics are not prefixes or equivalents of corrected trajectories.

Stress contract

`REF_stress` is an ASE six-component stress vector in eV/Angstrom^3 with order

```
xx yy zz yz xz xy
```

and ASE's stress sign convention. Stress and virial are never conflated. A missing stress label is represented through zero stress loss weight, not by a fabricated physical value.

Atomic-reference mapping

DATA7 `AtomicReferenceFitRecord` values are serialized into the target head as an explicit mapping from atomic number to energy. Conceptual record names are never placed in the MACE `E0s` field. The target head also carries an explicit head-local `atomic_numbers` literal containing only elements present in target configurations. The top-level `atomic_numbers` remains the union of target and replay elements for model construction. This distinction is required by MACE 0.3.16: without the head-local table, the target head inherits replay-only elements and incorrectly demands target E0 values for them. The fit-record digest remains in the job manifest and `TrainingProtocolIdentity`.

`ReplayPreparationPlan`

Supported modes are:

- `NONE`;
- `PRESELECTED` local replay;
- `EXTERNAL_TRUE_LABEL` local replay;
- `EXTERNAL_PSEUDOLABEL` local replay;
- `MP_SHORTCUT` preparation-only planning.

Fixed-file execution requires local replay train and monitor artifacts. `MP_SHORTCUT` is rejected at job-bundle construction until its selected replay file has been materialized and inspected.

Replay train and monitor configurations must be geometry-disjoint. The monitor is never used for gradients and later provides retention evidence. Each file records its SHA-256 digest, frame count, element set, geometry identities, and property keys.

`ReplayRetentionPolicy`

Declares the retention metric, maximum tolerated degradation, disjoint-monitor requirement, and failure behavior. DATA8 serializes this policy; DATA9 evaluates it against saved checkpoints.

FoundationCheckpointIdentity

Binds protocol artifacts to the exact foundation checkpoint path, file digest, model label, and optional model metadata. A changed checkpoint creates a new training protocol.

TrainingProtocolIdentity

Binds all choices that can change optimization or interpretation:

- target label domain;
- naive or multi-head replay mode;
- foundation checkpoint;
- replay-plan digest;
- DATA7 objective, weight, E0-fit, and checkpoint-policy digests;
- MACE compatibility lock and source probe;
- optimizer and precision settings;
- checkpoint-control policy;
- exposure backend;
- loader dry-run realization;
- selected training level;
- random seed.

A naive protocol and a replay protocol are different identities even if their target XYZ files are identical.

SealedEvaluationArtifact

Records locked interpolation-test membership and lineage without writing a test XYZ file. The artifact is explicitly unmaterialized. DATA9 may activate it only after a `ProtocolFreezeRecord` exists.

MaceJobArtifact

Represents one final-development job or one independent cross-validation fold. It binds:

- target training and checkpoint-monitor artifacts;
- optional replay train and monitor artifacts;
- explicit fold evaluation artifact excluded from the training configuration;
- YAML configuration;
- run script;
- protocol identity;
- loader dry-run record;
- file checksums and manifests.

Data8PreparationBundle

Collects all final and fold jobs for exactly one target label domain, plus local replay artifacts and sealed outer evaluation metadata. Multiple incompatible target label domains require separate DATA8 bundles.

Extended-XYZ and sidecar split

Extended XYZ carries only training labels, weights, and a stable `frame_uid`. Complete provenance remains in a canonical JSON sidecar keyed by frame UID:

- source occurrence and source-content identities;
- source frame index;
- composition and condition;
- geometry and label-payload identities;
- eligibility and selection lineage;
- weight and E0-fit records;
- file and policy digests.

This prevents long provenance payloads from becoming fragile XYZ header text.

Job layout

A bundle has the conceptual layout:

```
data8_bundle/
  shared/
    replay/replay_train.xyz
    replay/replay_monitor.xyz
  jobs/
    final/
      target_train.xyz
      target_monitor.xyz
      mace_config.yaml
      run_mace.sh
      job_manifest.json
    fold_00/
      target_train.xyz
      target_monitor.xyz
      fold_evaluation.xyz
      mace_config.yaml
      run_mace.sh
      job_manifest.json
    ...
  data8_bundle.json
```

`fold_evaluation.xyz` is an evaluation artifact only. Its path must not appear in `mace_config.yaml`. No locked interpolation-test XYZ exists in this tree.

MACE configuration contract

The target head contains explicit property keys and E0 values. A replay job also declares `pt_train_file` and `pt_valid_file`. The generated configuration shall include:

- `foundation_model`;
- `multiheads_finetuning`;
- target head and optional `pt_head` definitions;
- explicit energy, force, and stress keys;
- target/replay head weights;
- DATA7 global property weights;
- optimizer, seed, device, and floating-point precision;
- `save_all_checkpoints: true`;
- target-last native checkpoint-control settings;
- `real_pt_data_ratio_threshold: 0.0` by default;
- no test path.

The first adapter supports fixed-file training only. `CUSTOM_EPOCH_RESAMPLE` and `MULTI_JOB_RESAMPLE` remain later backends and cannot be represented merely by writing one static YAML file.

Cross-validation jobs

Each DATA5 fold produces an independent job:

1. DATA7 features, E0 values, weights, and selection are fit only on the fold's gradient-training domain.
2. The nested checkpoint monitor is exported as the MACE validation file.
3. The held-out evaluation fold is exported separately and excluded from the MACE configuration.
4. Replay artifacts and optimizer/checkpoint rules match the final protocol.
5. DATA9 trains a fresh model and evaluates the held-out fold only after the checkpoint decision.

A cross-validation family that omits replay cannot validate a final replay protocol.

Failure rules

DATA8 fails closed when:

- the MACE source probe does not match the active version lock;
- more than one target label domain enters one bundle;
- a DATA7 fold is missing or has incompatible DATA5 lineage;
- the foundation checkpoint cannot be hashed;
- an E0 mapping is incomplete for the target elements;
- a target label or training weight is unavailable;
- replay train and monitor overlap;
- local replay is incomplete;
- `MP_SHORTCUT` is passed to fixed-file execution;
- a locked-test or fold-evaluation path enters a training configuration;
- an extended-XYZ round trip changes labels, stress, weights, or frame order;
- serialized digests or file checksums fail.

Non-goals

DATA8 does not execute MACE, inspect actual checkpoint files, enforce replay retention numerically, aggregate out-of-fold results, choose a final model, construct a committee, freeze a protocol, activate locked tests, calibrate uncertainty, or acquire active-learning labels.

References

1. MACE documentation, "Multihead fine-tuning," official project documentation, accessed 2026-07-28.
2. MACE documentation, "Training," official project documentation, accessed 2026-07-28.
3. ACEsuit/MACE v0.3.16 source, `mace/cli/run_train.py`, official tagged source.
4. ACEsuit/MACE v0.3.16 source, `mace/tools/train.py`, official tagged source.
5. ACEsuit/MACE v0.3.16 source, `mace/tools/multihead_tools.py`, official tagged source.
6. ASE documentation, extended XYZ and stress conventions, version 3.29.0.

DATA9A hardening amendments

The production adapter applies the following stricter contracts:

- foundation fine-tuning requires a checkpoint-bound `foundation_residual` atomic-reference fit; direct total-energy E0 fitting is a from-scratch-only fallback;
- the foundation checkpoint and local replay files are staged under `shared/`, YAML paths are relative, and run scripts change to their own directory;
- extended-XYZ verification compares species/order, PBC, cell, positions, energy, forces, stress, config type, and all weights numerically;
- replay inspection rejects nonfinite/misshaped labels and internal exact duplicates, records stress coverage, and binds pseudo-labels to a checkpoint;
- top-level `atomic_numbers` is the union of target and replay elements;
- `heads.target_head.atomic_numbers` is the target-only element set, preventing replay-only species from becoming target-head E0 requirements;
- one explicit DATA7 ladder size is bound into every protocol identity.

These amendments are implemented in 0.20.37a0 as part of DATA9A hardening.

DATA9A2 executable-serialization amendments

The v0.3.16 runtime contract is stricter than a native YAML interpretation. `atomic_numbers`, `heads`, every nested head `atomic_numbers` value, and every nested head `E0s` mapping are emitted as scalar strings containing deterministic Python literals. DATA8 does not emit unsupported `weight_pt` or `weight_ft` options.

The loss name recorded in this amendment was the lowercase parser choice `universal`. That is historical: the retired DATA8 preparation topology is no longer reachable from any current command, and the current executable loss family is `stress` (see "Executable loss family and weighting layers" below).

For preselected fixed-file replay, target and replay training exposure is realized by multiplying each training structure's extended-XYZ `config_weight` by the corresponding target or replay head scale. Validation, monitor, fold-evaluation, and locked-test structures remain unscaled. The target sidecar records base, scale, and realized weights. These amendments are implemented in 0.20.39a0 and are qualified by the real-MACE DATA9A2 records.

Selectable fine-tuning precision

`MaceOptimizerPolicy.default_dtype` SHALL be either `float32` or `float64` and SHALL be serialized into every generated MACE configuration. Because the optimizer policy is part of `TrainingProtocolIdentity`, changing precision creates a different protocol even when every dataset artifact is unchanged.

The MPA-0 foundation checkpoint may remain uniformly float64. The DATA8 bundle SHALL reference it unchanged; runtime conversion to float32 is requested through MACE's `default_dtype` option. DATA8 SHALL NOT claim the output precision from configuration text alone. DATA9A runtime realization must inspect the saved model and target-head model and prove that all floating parameters and buffers are uniformly the requested dtype.

DATA9A9b production orchestration

DATA9A9b calls the native DATA8 builder only after all planned final and fold-local DATA7 bundles verify. The production plan binds the foundation checkpoint, MACE compatibility probe, optimizer/checkpoint/export policies, selection size, and exact replay train/monitor plan before artifact generation. A `ProductionData8ArtifactRecord` binds the native DATA8 bundle digest and every relative file path and SHA-256 in the emitted tree. A partial or modified tree is not accepted as production materialization evidence.